Yahya Masri

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Education

George Mason University Fairfax, VA

B.S. in ***Computer Science*** | ***GPA: 3.92*** *May 2027*

- Courses: Data Structures & Algorithms, Concurrent & Distributed Systems, Computer Systems and Programming, Software Engineering, Natural Language Processing

Experience

U.S. Naval Research Laboratory (NRL) Sep 2026 – Present

Student Researcher Co-op *Washington, D.C.*

- Built a neuro-symbolic RL pipeline integrating natural-language-to-LTLf translation with PPO agents
- Achieved 92% capture success across 150 held-out simulations through temporal guidance and reward tuning

NSF Spatiotemporal Innovation Center, GMU Feb 2024 – Present

Research Assistant *Fairfax, VA*

- Built an LLM-enabled digital twin for a 600+ machine data center, integrating Prometheus/Loki telemetry, PostgreSQL, and NVIDIA Omniverse through 19 MCP tools for natural-language diagnostics and 3D control.
- Published 5 peer-reviewed publications, including 3 as first author, on LLM/SLM evaluation, RAG, digital twins, and AI-driven conflict analysis.
- Built a real-time conflict intelligence system using LLMs to extract, validate, and geocode 3,543 conflict events from 9,353 news articles, streaming results to an interactive Next.js/MapLibre dashboard.

SAP NS2 May 2026 – August 2026

Software Engineer Intern *Herndon, VA*

- Built an AI usage analytics agent to transform 400K+ internal AI interactions across 847 users into leadership analytics, enriching AWS OpenSearch data with business-line, use-case, token, and time-savings metadata.
- Increased enrichment throughput by 5.6x, from 25 to 140 records/min, by replacing page-synchronous processing with continuous work scheduling across 12 concurrent workers and horizontally scaled instances.
- Secured a CUI-authorized internal AI platform with role-scoped access controls below the LLM, using adversarial red-team testing to uncover and remediate identity-leakage paths.
- Placed 1st overall at an internal among 300 participants by building an agentic workspace using live Microsoft Graph API data to surface meetings, priorities, and follow-ups.

Lillup Jan 2026 – Mar 2026

ML Engineer Intern *Remote*

- Built a local CPU-based RAG pipeline using ChromaDB and llama-cpp-python achieving 90% source-grounded responses at 2s query latency.
- Cut time-to-first-audio from 7s to 500ms (14x) on CPU by streaming LLM output sentence-by-sentence into TTS and switching from a 3B Llama model to Qwen3-0.6B for faster on-device inference.

Projects

Autonomous FPV Drone Racing System | *Anduril — AI Grand Prix* | GitHub Jul 2026

- Built an autonomous FPV drone-racing system combining computer vision and PPO reinforcement learning
- Engineered a Gymnasium/PPO training pipeline with reward shaping and deterministic evaluation
- Optimized real-time gate detection to 1.5ms per frame using HSV segmentation and contour filtering

Autonomous Solana Trading Agent | *Winner – Best Use of Solana Track* | GitHub Feb 2026

- Placed 1st in the Best Use of Solana track out of 500+ participants at HackFax X PatriotHacks '26
- Deployed an agentic trading system on AWS EC2 using OpenClaw and Bedrock for autonomous paper trading
- Built server-side trade execution and risk management APIs to block 95% of high-risk trades before execution.
- Developed risk and strategy APIs to validate AI-generated trades, log risk events, and persist execution history

Technical Skills

Languages: Python, Java, C, SQL, Go
AI/ML & Frameworks: PyTorch, FastAPI, LangChain, MCP, RAG, HuggingFace, ChromaDB, FAISS, scikit-learn
Cloud & Infrastructure: AWS, Docker, Kubernetes, Linux, OpenSearch, PostgreSQL, MongoDB, Prometheus, Grafana, Git